PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Care Interventions
AUTHORS	Rizvi Jafree, Sara; Mahmood, Qaisar; Mujahid, Sohail; Asim, Muhammad; Barlow, Jane

VERSION 1 – REVIEW

REVIEWER	Hussain, Basharat University of Nottingham, Nottingham University Business School
REVIEW RETURNED	22-Feb-2022

GENERAL COMMENTS	Comments to the Author The author(s) report A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Level Interventions. Overall, the review is much needed and findings will be helpful for planning further interventions in this area. Please see the comments below to help strengthen the review: Introduction: This part is written well and offers very good overview of women's health issues and health service delivery arrangements in Pakistan. However, this section can be strengthen by adding some literature on critical role primary level healthcare can play in achieving good health for women in Pakistan. I will also suggest that the author(s) consider highlighting types of healthcare needs women in Pakistan will meet at primary healthcare level. Methods This section needs major revision: 1. Why a narrative synthesis is a better option when focus of the review is only on RCTs and quasi-experimental design. 2. There is discrepancy in the stated review objectives in the published protocol and in the actual review undertaken. Need explanation. 3. Searches undertaken timeframe is also given differently in the published protocol and actual review. Explanation is needed. 4. Initial screening – How many articles were included at this stage. Were there any differences among the reviewers in the initial screening? It appears that the lead author screened titles only and then handed over to second reviewer to review abstract of the included titles. 5. Data extraction: Focus of the review is on primary care intervention, so it would be helpful to include intervention delivery method in the data extraction as well. Also data analysis technique used and reference where the paper is published. 6. Included studies : 135,553 references are found in the searches. This number is very high compare to duplicates and included studies (18). Usually the difference is 20-30%. So need checking if the searches are formulated correctly, relevant filters

	are applied (e.g. Pakistan). Support from an Information Specialist is usually advised for this step of a review. Results: Need definition/description of three 'themes' before presenting results from the review. Discussion section is well written The strength and limitation section is well written

REVIEWER	Caglia, Jacquelyn Merck & Co Inc, Social Business Innovation
REVIEW RETURNED	12-Mar-2022

GENERAL COMMENTS	Pg 2, line 50 - "promote effective working"; consider rephrasing Pg 2, line 60 - consider defining RCT at first use Pg 3, line 79 - I'm not sure I understand the point here about using patient-reported outcomes for measurement. Are the authors trying to say that some studies in the review used patient-reported outcomes only vs. updated clinical information? Pg 3, lines 90-92 - consider referencing how Pakistan fares with regards to the Sustainable Development Goals, given that the MDG framework referenced was completed in 2015 Pg 13, line 247 - consider defining CBT at first use Overall, I think that this review adds to the literature base about effective primary care interventions to address women's health needs in Pakistan. Most important is the conclusion that more work is needed to better understand effectiveness, especially with more horizontal programs that touch upon a variety of needs across the whole of women's health.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
R5. The author(s) report A Narrative Synthesis Systematic Review of Pakistani Women's Health Outcomes from Primary Level Interventions. Overall, the review is much needed and findings will be helpful for planning further interventions in this area. Please see the comments below to help strengthen the review:
Author Response: Thank you so much for your time and valuable suggestions.

R6. Introduction:
This part is written well and offers very good overview of women's health issues and health service delivery arrangements in Pakistan. However, this section can be strengthen by adding some literature on critical role primary level healthcare can play in achieving good health for women in Pakistan. I will also suggest that the author(s) consider highlighting types of healthcare needs women in Pakistan will meet at primary healthcare level.
Author Response: Thank you for this suggestion, we have added more information pertaining to: (i) the critical role primary level healthcare can play in achieving good health for women in Pakistan, and (ii) types of healthcare needed by women in Pakistan at primary healthcare level (Lines: 85-144):
Creating an effective primary care system for women has not been a priority in Pakistan1. One of the difficulties is that a model for an effective primary healthcare setup, for women of different age groups and across the life course has not been agreed upon globally2, or within Pakistan, to date3. Research suggests that for a comprehensive primary healthcare system for women to be effective in developing regions, service delivery must be led by state efforts and public health policy4. There is also

agreement that a primary healthcare model must be developed regionally, based on a nations historical background, health policies, revenue system, and cultural beliefs, and based on a review of existing services5.

Despite the inadequate primary healthcare services and low utilization6, the majority of Pakistani women need to be provided state support for primary healthcare due to critical socio-economic problems such as poverty and unemployment 7, or informal sector employment which excludes women from employer health protection 8. There is a need to further support women at the primary care level to reduce maternal mortality 9, chronic disease 10, infectious diseases 11, injury and violence 12, mental health problems 13, and health issues related to aging 14. The country also has a number of unmet Sustainable Development Goals, with key issues such as high maternal mortality rates and the absence of universal healthcare services for women15.

Primary healthcare in Pakistan

Since 2011 responsibility for the delivery of Pakistan's healthcare system has been devolved to the provinces. The Federal Ministry of National Health Service Coordination and Regulation is responsible for monitoring and allocation of the budget to the four provinces of the country including Baluchistan, Khyber Pakhtunkhwa, Sindh and Punjab. The overall framework and structure for the delivery of primary healthcare services is the same across the country, with services including: reproductive and maternal health; vaccination and immunisation; health awareness; referral to secondary and tertiary sectors; and routine counseling. However, the outreach and implementation varies according to provincial development, with Punjab and KPK provinces performing better, compared to Baluchistan, Sindh and Gilgit-Baltistan 16.

The current state-provided universal healthcare system in the country, the Sehat Sahulat Program, primarily focuses on payment for hospitalization, with a capping of USD 1,906 per person17. This program is severely limited due to low outreach of less than 0.5% of the national population, and is only accessible to women with a national identity card, excluding thereby the majority of disadvantaged women in the country17. Pakistan has a functional primary care setup for women, called the Lady health Workers (LHW) Programme, which was launched by the government of Pakistan in 1994, and delivers door-to-door services to women at their homes18. LHWs provide key services for immunization, family planning, counseling and maternal health services. In addition, there are community midwives (CMWs) trained and then deployed to practice privately in the community for safe delivery services19. Primary care services are also provided within the community at basic health units (BHUs) and rural health centers (RHCs), which are allocated a physician or medical officer and a nurse, apart from the LHW. The BHUs and RHCs aim to provide clinical consultancy and basic reproductive health services.

Existing evidence

While a number of studies have reported that the LHW programme and primary setup for maternal healthcare has been successful in improving family planning 26 27 and mortality rates 28, others have reported that primary health care services have not yielded positive outcomes for women's health in Pakistan 15 29 30. This reflects the fact that access and outreach remains low for the 110 million women of Pakistan, primarily due to the inadequate ratio of primary care services with regard to LHW/CMW to population ratios; BHU and RHC doctor to population ratios; and bed to population ratios for RHCs 20 21. The LHW and CMW programs have also been criticised in terms of the limited services for family planning24. During the recent pandemic the LHWs were additionally assigned responsibility for promoting infection preparedness and prevention, without any additional compensation or staff allocation, contributing to inefficiencies 25.

Similarly, the BHU and RHC services have also been criticized due to low quality services, inadequate budget allocation, and poor coordination from the center 22. There is also low utilization and visitation to BHUs and RHCs by women due to patriarchal traditions preventing mobility, mistrust, and distance23. To date there has been no systematic review of rigorous evaluations to identify which primary care interventions are effective in improving women's health outcomes in Pakistan. The aim of this review was to systematically review the available literature for studies that have evaluated the effectiveness of primary healthcare services for women, in order to inform future health research and

policy.

R7. Why a narrative synthesis is a better option when focus of the review is only on RCTs and quasi-experimental design.
Author Response: Thank you. We have explained in the manuscript (Lines 212-4):
Due to the high level of heterogeneity across studies in terms of interventions and outcomes, a narrative summary of the findings was necessary.

And also in Lines 177-8:
The final 18 included articles had a diverse range of interventions and outcomes, which meant that it was only possible to provide a narrative synthesis32.

R8. There is discrepancy in the stated review objectives in the published protocol and in the actual review undertaken. Need explanation.
Author Response: Thank you. We have added the following explanation in the text (Lines 179-85):
Although the protocol refers to two objectives (The primary objective is to identify the effectiveness of the intervention in terms of how successful it was in improving health of women; whereas the secondary aim is to identify barriers and facilitators for delivery of primary healthcare services) this paper only reports the findings of the first of these. The review pertaining to the secondary aim referred to in the protocol paper, will be submitted as a separate manuscript entitled: barriers and facilitators to the delivery of primary health care services for women in Pakistan.

R9. Searches undertaken timeframe is also given differently in the published protocol and actual review. Explanation is needed.
Author Response: Thank you. We have added this information (Lines 153-6):
We were unable to start searching during the dates mentioned in the protocol (January to June 2021) due to a combination of factors such as investigator time constraints, transfer to online teaching for faculty, and COVID-19 social distancing making it difficult for co-authors to meet and coordinate.

R10. Initial screening – How many articles were included at this stage. Were there any differences among the reviewers in the initial screening? It appears that the lead author screened titles only and then handed over to second reviewer to review abstract of the included titles.
Author Response: Thank you. We have clarified this and included more information (Lines 171-8).
An initial screening of studies based on titles, was performed by the first author for 109,841 studies. An independent screening was also done by the second author. Discussions about which articles met the inclusion criterion were held. At second step, the abstracts of 543 studies were reviewed by the first and third authors independently. The main reasons for exclusion were that articles were not interventions in Pakistan or exclusively in Pakistan, did not relate to health interventions or primary health care outcomes for women, were protocol or feasibility studies, or did not have a control group. The final 18 included articles had a diverse nature of interventions and assessment of different outcomes, which meant that it was only possible to provide a narrative synthesis32.

R11. Data extraction: Focus of the review is on primary care intervention, so it would be helpful to include intervention delivery method in the data extraction as well. Also data analysis technique used and reference where the paper is published.
Author Response: Thank you so much for this suggestion. We have included this information, (Lines 187-91):
The following information was abstracted by the first author from the interventions that met our eligibility criteria in terms of targeting women's health outcomes: study design; setting; participant type; sample size; aim; type of intervention; intervention delivery method and description; outcomes being measured; data analysis technique; results; and complete citation of the study. Extraction of the data was reviewed for consistency by the third author.

R12. Included studies: 135,553 references are found in the searches. This number is very high compare to duplicates and included studies (18). Usually the difference is 20-30%. So need checking if the searches are formulated correctly, relevant filters are applied (e.g. Pakistan). Support from an Information Specialist is usually advised for this step of a review.
Author Response: Thank you. Yes, 135,553 references were found. The details of the search terms and references found against selected databases have been shared in Supplementary File No 1. The search strategy was designed with the support of FCCUs, Chief librarian Mr. Farrukh Shahzad (furrukhshahzad@fccollege.edu.pk) and supervisor of this project. A total of five search terms were developed and, and in each search term, the word Pakistan was entered. The search results have been reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart, presented in Figure 1. A total of 543 studies were retrieved and assessed against the selection criteria, after reviewing the abstract of those studies. Since this review was focused on the effectiveness of primary healthcare interventions for women's health outcomes, a total of 18 studies were included out of 543 (other studies assessing barriers and facilitators , the secondary aim of this review project, also explained above, have been separated and will be submitted in another manuscript.

R13. Need definition/description of three 'themes' before presenting results from the review.
Author Response: Thank you. The definition and description of the four themes in Results section has been added (Lines 256-66):
Due to the heterogeneity in terms of the included study designs and interventions, a narrative report of effectiveness has been provided, organized according to the type of intervention as follows: : A. Counseling Interventions- including one-to-one or group sessions to improve mental health or family planning practices; B. Health Education and Awareness Interventions- including group or one-on-one educational sessions to TBAs and women of reproductive years about safe maternal practices or malaria prevention; C. Social and Psychosocial Interventions- including group social support sessions (peer-support, behavioral activation, and problem solving) and one-on-one mental health support; and D. Combination Interventions- Group or one-on-one sessions for family planning and contraception use or wider maternal health indicators using a combination of different approaches such as educational outreach, clinical assessment, transport provision and referral.

R14. Discussion section is well written
Author Response: Thank you so much.

R15. The strength and limitation section is well written
Author Response: Thank you again.

Reviewer: 2
R16. Pg 2, line 50 - "promote effective working"; consider rephrasing
Author Response: Thank you. We have rephrased this as: "…promote effectiveness for women's health outcomes." (Line: 51).

R17. Pg 2, line 60 - consider defining RCT at first use
Author Response: Thank you. We have defined it as first use (Lines: 61 and 167).

R18. Pg 3, line 79 - I'm not sure I understand the point here about using patient-reported outcomes for measurement. Are the authors trying to say that some studies in the review used patient-reported outcomes only vs. updated clinical information?
Author Response: Yes, thank you, we mean: "some studies in the review used patient-reported outcomes only and thereby no objective clinical data?" We have now edited this in the manuscript (Lines 80-1).

R19. Pg 3, lines 90-92 - consider referencing how Pakistan fares with regards to the Sustainable Development Goals, given that the MDG framework referenced was completed in 2015
Author Response: Thank you so much for pointing this out. The SDGs have now been mentioned and referenced accordingly (Lines 99-101):
The country also has a number of unmet Sustainable Development Goals, with key issues such as high maternal mortality rates and the absence of universal healthcare services for women15.

R20. Pg 13, line 247 - consider defining CBT at first use
Author Response: Thank you. This has been addressed above (Lines 304-5).

R21. Overall, I think that this review adds to the literature base about effective primary care interventions to address women's health needs in Pakistan. Most important is the conclusion that more work is needed to better understand effectiveness, especially with more horizontal programs that touch upon a variety of needs across the whole of women's health.
Author Response: Thank you so much for your support, time, and review.

VERSION 2 – REVIEW

REVIEWER	Hussain, Basharat University of Nottingham, Nottingham University Business School
REVIEW RETURNED	21-Jun-2022

GENERAL COMMENTS	Authors have addressed the points raised. I recommend its publication.